# Detecting Subjectivity in Textual Reviews

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# CERTIFICATE

This is to certify that the work contained in the project entitled **”Detecting Subjectivity in Textual Reviews”** is a bonafide work of **Jakadeer.V** (Roll No. 142201023) and **Ashwini Kannan.K** (Roll No. 142201007), carried out in the Department of Data Science and Engineering, Indian Institute of Technology Palakkad under my guidance and that it has not been submitted elsewhere for a degree.

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# Abstract

Subjective bias is an inherent component of textual reviews, shaped by psychological tendencies, linguistic salience, and individual reviewer preferences. Building on insights from behavioral economics, cognitive linguistics, and recent work on subjective interventions in reviews, this study develops a systematic framework for detecting and analyzing subjective bias in movie review text. We introduce a nine-category taxonomy capturing core cognitive bias types—Expectation Bias, Comparative Lens, Representation Critique, Omission Focus, Moral/Political Projection, Selective Emphasis, Performance Appraisal, Authenticity Bias, and Disposition Bias.

To operationalize this taxonomy at scale, we design a deterministic prompt-engineering pipeline that constrains large language models to assign exactly one dominant bias per sentence using salience-driven lexical cues and priority-ordered decision rules. Applying this framework to thousands of IMDB review sentences, we construct reviewer-level bias profiles and analyze them using Principal Component Analysis and Factor Analysis.

Results reveal stable latent dimensions of reviewer subjectivity, identifying clusters of highly subjective, emotionally driven, and balanced reviewers. Our findings demonstrate that subjective bias in textual reviews is not random noise but a structured and measurable phenomenon, and that large language models—guided by cognitively grounded prompts—can effectively capture these underlying bias patterns.

# Contents

# Chapter 1

# Introduction

Human evaluations in textual reviews are far from objective. Reviewers do not simply describe what they observe; instead, they filter their judgments through personal expectations, prior beliefs, social norms, and cognitive shortcuts. This subjectivity is not random—psychological research shows it follows systematic and predictable patterns. Dan Ariely's *Predictably Irrational* demonstrates that human decisions and evaluations routinely deviate from rational models, shaped by anchoring, expectation mismatches, emotional coherence, and relativity-based comparison (1). These biases influence how individuals interpret media and express opinions in review text.

From a linguistic perspective, Giora's Graded Salience Hypothesis (GSH) provides further explanation for how such biases surface in language (2). The theory asserts that the most salient, familiar, and conventional meanings of words are accessed first, regardless of context. This offers a strong theoretical basis for bias detection, as reviewers tend to express expectations ("I expected more"), comparisons ("not as good as the original"), realism concerns ("unbelievable plot twist"), or moral judgments ("pushes an agenda") using highly salient and easily identifiable linguistic cues. These cues allow computational models to treat subjective bias as a measurable linguistic phenomenon.

The NLP literature reinforces this perspective. Recent research highlights that reviewers frequently introduce subjective interventions—explicit or implicit linguistic signals that reflect personal stance, bias, or perspectival shading (3). This emphasizes that subjective interventions are both pervasive and detectable, provided that systems account for the inherent ambiguity, annotator disagreement, and interpretive variability of review text. It further argues that automatic systems can successfully learn interpretable patterns of reviewer subjectivity when guided by structured annotation frameworks.

## 1.1 Motivation and Problem Statement

Motivated by these insights, this project aims to develop a systematic methodology for detecting and classifying subjective biases in movie reviews at the sentence level. Rather than treating subjectivity as open-ended or purely qualitative, we define a structured taxonomy of nine bias categories—including Expectation Bias, Comparative Lens, Representation Critique, Omission Focus, Moral/Political Projection, Selective Emphasis, Performance Appraisal, Authenticity Bias, and Disposition Bias. Each category captures a distinct psychological tendency and its corresponding linguistic realization.

To operationalize this taxonomy, we design a prompt engineering pipeline that constrains a large language model to identify exactly one dominant bias per sentence. The prompt incorporates salience-based reasoning, lexical cue identification, deterministic tie-breaking, and transparent justification rules. Using this framework, both OpenAI and local Ollama models annotate thousands of IMDB-scraped review sentences.

## 1.2 Objectives and Contributions

The resulting bias labels are aggregated into reviewer-level bias profiles, forming a structured bias matrix. Statistical techniques such as Principal Component Analysis (PCA) and Factor Analysis (FA) are then applied to uncover latent dimensions of reviewer behavior. These analyses reveal clusters of highly subjective reviewers, balanced reviewers, and those with strong emotional or evaluative tendencies—echoing findings from both psychological research and the subjective intervention detection literature.

Overall, this project makes three key contributions:

1. A cognitively grounded, linguistically informed taxonomy of subjective biases in media reviews.

2. A deterministic prompt engineering framework enabling scalable, reproducible LLM-based bias annotation.

3. A reviewer-level analytical methodology that uncovers latent subjectivity structures through PCA and factor-analytic modeling.

By integrating cognitive psychology, linguistic theory, and modern language models, this work demonstrates that subjective bias in textual reviews is not merely noise—it is a structured, analyzable phenomenon that can be systematically detected, classified, and interpreted.

# Chapter 2

# Background and Related Work

## 2.1 Understanding Subjectivity

Subjectivity refers to the presence of personal opinions, beliefs, expectations, emotional judgments, or biases within linguistic expression. Unlike objective statements, subjective language is not necessarily verifiable, as it depends on an individual's internal reasoning, values, and experiences. In natural language processing (NLP), subjectivity has traditionally been examined through sentiment analysis and opinion mining, where the focus is on polarity (positive/negative) rather than the cognitive mechanisms behind judgments (4).

However, existing research highlights that subjectivity is multidimensional, involving cognitive, evaluative, and socio-cultural cues that extend far beyond simple emotions. Recent work in stance detection, pragmatic inference, and narrative psychology suggests that subjective expressions include latent structures of reasoning, often tied to psychological tendencies such as projection, comparison, expectation, moral judgement, or prior belief formation.

Thus, subjectivity is increasingly understood not merely as sentiment but as a psychologically grounded process, observable through linguistic patterns and measurable via computational models. Our work follows this direction by framing subjectivity as a set of cognitive interventions—detectable not only through lexical markers but also through reviewer logic and justifications.

## 2.2 Why Movie Reviews?

Movie reviews offer a uniquely rich domain for studying subjectivity due to the characteristics summarized in Table 2.1.

Unlike product reviews or political comments, movie reviews often include long-form reflective reasoning, enabling the extraction of subtle psychological biases.

Table 2.1: Justification for using movie reviews as the study domain

| **Reason** | **Justification** |
|---|---|
| Opinion-rich domain | Reviews are inherently subjective and encourage personal interpretation. |
| Narrative interpretation | Films trigger reflections on morality, science, history, identity, or realism. |
| Diverse reviewer backgrounds | Reviewers come from different knowledge domains and value systems. |
| Depth beyond sentiment | Reviews include argumentation, comparison, justification, and projection. |
| Availability of data | Large public datasets allow scalable analysis across genres and reviewers. |

A single reviewer may simultaneously express disappointment, comparison with other works, moral judgments, or technical evaluation—making movie reviews an ideal space to explore how subjective bias manifests linguistically.

## 2.3 Gap in Existing Literature

Prior studies primarily focus on sentiment polarity or feature-based evaluation (e.g., acting, visuals, direction) (5). However, they do not examine the cognitive nature of subjective interventions. Most existing work classifies review content into positive/negative opinions or topic-based categories. Few studies attempt to characterize the reasoning behind reviewer judgments. The lack of a principled taxonomy for subjective bias limits our ability to detect the type of subjectivity present in reviews.

Therefore, this study aims to:

- Move beyond sentiment classification toward deeper subjective modeling.
- Identify latent psychological constructs embedded in review sentences.
- Demonstrate that LLM-based annotation and statistical modelling can approximate cognitive patterns in human judgments.

# Chapter 3

# Subjective Bias Taxonomy

Based on our analysis of diverse movie reviews, we identified nine distinct subjective biases that frequently appear as subjective interventions in textual reviews. These biases provide a structured framework to quantify and categorize subjectivity, enabling us to detect how reviewers impose their personal beliefs, expectations, or preferences on films.

## 3.1 Definitions of Subjective Biases

### 3.1.1 Expectation Bias

**Definition:** This occurs when a reviewer judges a movie based on what they expected it to be rather than evaluating what it actually was. Their dissatisfaction arises from unmet personal expectations—not necessarily flaws in the film itself.

**Example:** A reviewer criticizes *Oppenheimer* for lacking physics and scientific details, stating that it focused too much on politics instead of the bomb or its consequences.

### 3.1.2 Comparative Lens

**Definition:** The reviewer evaluates a movie by comparing it to another film, book, or reference material, often implying that it did not live up to the original.

**Example:** A review of *Oppenheimer* compares it to the book *American Prometheus*, stating the film fails to do justice to the source material.

### 3.1.3 Representation Critique

**Definition:** The reviewer questions how accurately certain characters, groups, or perspectives are represented in the film. This often includes complaints about stereotypes, imbalance, or underrepresentation.

**Example:** In a review, the casting choice for Jane Hawking is criticized as unrealistic, making her relationship with Stephen Hawking difficult to believe.

### 3.1.4 Omission Focus

**Definition:** The reviewer highlights important topics or details that were left out of the film, implying that these omissions weakened the storytelling or accuracy.

**Example:** A reviewer expected more details about the Manhattan Project and other historical elements that were not shown in the film.

### 3.1.5 Moral / Political Projection

**Definition:** When a reviewer inserts their own moral or political viewpoint into the evaluation of the film, judging it based on ideological alignment rather than cinematic quality.

**Example:** A review claims that the important issue in *Oppenheimer* is whether the protagonist was a communist—projecting a political angle onto the narrative.

### 3.1.6 Selective Emphasis Bias

**Definition:** The reviewer focuses intensely on one specific element (positive or negative) of the movie, while ignoring the rest of the film's content and context.

**Example:** A reviewer relates the protagonist's struggles to their own mental health experiences, emphasizing only that aspect while overlooking other elements of the film.

### 3.1.7 Performance Appraisal

**Definition:** Focuses on actor performance, delivery, or mannerisms—evaluating the film primarily through character portrayal and acting quality.

**Example:** Benedict Cumberbatch was criticized for having the same persona as Sherlock, making the performance feel repetitive.

### 3.1.8 Authenticity Bias

**Definition:** The film is judged on technical or factual accuracy, often based on domain knowledge—especially when related to history, science, or expertise.

**Example:** A reviewer, with a computer science background, claims that the portrayal of cryptography in *The Imitation Game* is inaccurate and implausible.

### 3.1.9 Disposition Bias

**Definition:** The reviewer's personal likes, dislikes, or preferences toward a theme or subject influence their judgment—regardless of cinematic quality.

**Example:** A reviewer admits disliking mathematics and expresses that films about mathematicians are unappealing to them, directly affecting their review.

## 3.2 Rationale for Selecting These Biases

The chosen taxonomy was not imposed arbitrarily; it emerged from a combination of data-driven analysis and cognitive reasoning. These nine biases were selected due to the following reasons:

### 3.2.1 Empirical Frequency in Real Reviews

We analyzed a wide range of reviews and found consistent patterns of subjectivity. The selected biases appeared repeatedly and naturally, making them reliable for annotation and modelling. They therefore represent real reviewer behavior, not theoretically imposed categories.

### 3.2.2 Coverage of Diverse Cognitive Processes

Each bias reflects a distinct mode of human judgment, covering emotional, ideological, comparative, technical, and evaluative reasoning. Together, they represent a holistic coverage of subjective interpretation as shown in Table 3.1.

Table 3.1: Cognitive coverage of bias taxonomy

| **Cognitive Orientation** | **Bias Categories** |
|---|---|
| Expectation & Prior Belief | Expectation Bias, Disposition Bias |
| Comparison-Based Reasoning | Comparative Lens |
| Social & Ideological Judgement | Moral/Political Projection, Representation Critique |
| Information Expectation | Omission Focus |
| Detail-Oriented Focus | Selective Emphasis Bias |
| Technical Accuracy | Authenticity Bias |
| Artistic Performance | Performance Appraisal |

### 3.2.3 Suitability for Computational Modelling

For any computational task, subjective categories must be distinct, labelable, and analyzable. These nine biases fulfill all three criteria:

- Distinct boundaries between biases
- LLM-promptable via clear definitions
- Statistically meaningful for PCA, factor analysis, and clustering

Therefore, they provide a practical foundation for detecting and quantifying subjectivity using both linguistic and statistical methods.

# Chapter 4

# Assumptions of the Study

## 4.1 Overview

The validity of our bias-annotation pipeline rests on several theoretical and methodological assumptions concerning linguistic processing, human cognition, and corpus behavior. These assumptions ensure that our framework remains coherent, operationally feasible, and interpretable. Beyond general cognitive principles, our assumptions are strongly supported by the Graded Salience Hypothesis (GSH) proposed by Giora, which argues that salient (familiar, conventional, prototypical) meanings are accessed first—regardless of context. This theory directly informs how we treat lexical cues, bias markers, and reviewer intent.

## 4.2 Core Assumptions

### 4.2.1 Sentence-Level Bias Singularity

We assume that each sentence contains one dominant bias, even if multiple interpretations are possible. According to the GSH, the most salient meaning or cue is activated first and strongest. Thus, even if a sentence contains layered meaning, the most salient lexical or semantic cue (e.g., *expected*, *better than*, *unrealistic*) tends to dominate cognitive interpretation. This aligns with our single-label design: the system selects the most salient bias expressed in the sentence.

### 4.2.2 Presence of Detectable Subjective Biases

We assume that media reviews inherently contain subjective biases that are linguistically detectable. Giora notes that salient meanings are typically encoded in familiar, conventionalized expressions. Reviewers often rely on such familiar evaluative frames—*expected more*, *not as good as*, *unrealistic*—making subjective bias measurable through surface-level language cues.

### 4.2.3 AI Approximation of Human Interpretation

We assume that large language models can approximate the salience-driven processing patterns humans use when interpreting bias. Because GSH shows that salient cues are processed automatically and robustly, LLMs trained on large corpora naturally prioritize the same salient features that humans attend to. Thus, models can reliably map reviewer statements to psychological bias categories.

### 4.2.4 Biases as Latent Psychological Constructs

We assume that biases in reviews arise from latent cognitive tendencies (e.g., expectation mismatch, comparative framing, moral evaluation). Giora emphasizes that context does not erase salience but modulates it; literal or figurative meaning is still grounded in underlying cognitive salience patterns. This supports our view that bias categories correspond to cognitively stable constructs expressed through predictable linguistic forms.

### 4.2.5 Reviewer-Level Aggregation Reflects Overall Bias

We assume that aggregating sentence-level labels yields a stable profile of a reviewer's overall bias tendencies. Giora's model suggests that salient meanings accumulate in discourse and continue to guide interpretation. Similarly, repeated activation of particular bias types across sentences reflects stable reviewer tendencies rather than random noise.

### 4.2.6 Corpus Representativeness

We assume that the review corpus reflects typical linguistic and cognitive patterns of media evaluation. Since the GSH is grounded in universal cognitive processing of salience, it supports using a representative dataset to generalize findings about bias expression.

### 4.2.7 Minimal Impact of Noise, Sarcasm, and Figurative Language

We assume that sarcasm, irony, and figurative expressions do not significantly distort automated bias detection. Giora explicitly notes that even in figurative or ironic contexts, salient literal meanings are activated first before reinterpretation occurs. Thus, although sarcasm may invert intent, the surface cue still carries salience, allowing our model to reliably extract the dominant linguistic signal.

# Chapter 5

# Dataset Collection

## 5.1 Source of Data

The primary data source for this study is IMDb user reviews, which provide rich, free-form textual feedback along with meta-information such as user ratings and helpfulness votes. IMDb reviews are particularly suitable because they are:

- User-generated and opinion-rich
- Accompanied by explicit star ratings (1–10)
- Associated with helpfulness metrics (likes/dislikes) that reflect community reception

For each selected movie, we collected reviews across the full rating spectrum (1–10 stars) to capture both highly critical and highly appreciative subjective perspectives.

## 5.2 Automated Scraping Procedure

Data collection was conducted using a custom Python script that combines `requests`, `BeautifulSoup`, and `Selenium` for robust extraction of both static and dynamically loaded content.

For a given movie, identified by its IMDb ID (e.g., `tt15398776` for *Oppenheimer*), the script iterates over rating filters from 1 to 10 and paginates through the corresponding review pages. For each (*movie*, *rating*) combination, it performs the following steps:

### 5.2.1 HTTP Request and HTML Parsing

- Sends a GET request to URLs of the form:
  `https://www.imdb.com/title/{movie_id}/reviews/?rating={rating_filter}&start`
- Uses a browser-like User-Agent string to reduce blocking and rate-limiting.
- Parses the returned HTML using BeautifulSoup to extract:
  - Movie title
  - Review title
  - Review body text
  - Displayed star rating (if present)

### 5.2.2 Dynamic Content via Selenium

- Some metadata, such as helpful "likes" and "dislikes" counts, is rendered dynamically.
- To access this, the same page is opened in a Selenium-controlled Chrome instance.
- After a short delay to allow the page to load, the page source is parsed again with BeautifulSoup to extract:
  - Number of users who found the review helpful (up-votes)
  - Number of users who did not find it helpful (down-votes)

### 5.2.3 Record Construction

For each review container detected on the page, the script constructs a record with the following fields:

- `movie` – movie title
- `rating` – numerical star rating
- `likes` – number of up-votes
- `dislikes` – number of down-votes
- `title` – user-provided review title
- `review` – full review text

These entries are appended to in-memory lists and later converted into a pandas DataFrame.

### 5.2.4 Export per Movie

For each movie ID, the collected reviews are saved to a CSV file of the form: `"{movie_id}_all_ratings_reviews.csv"`. This file serves as the raw review corpus for that specific movie. A short delay is inserted between page requests to be polite to the server and to reduce the risk of being rate-limited.

## 5.3 Corpus Construction Across Movies

The study examines subjectivity across multiple films rather than a single title. After collecting per-movie CSV files using the scraping procedure above, a separate preprocessing script was used to:

1. Load all per-movie CSVs into pandas DataFrames.
2. Concatenate them into a single unified corpus, preserving the `movie` field as an identifier.
3. Remove duplicates (e.g., if the same review was fetched more than once during pagination).
4. Standardize column types (e.g., converting `rating`, `likes`, and `dislikes` to numeric types).

The resulting corpus constitutes the base dataset for all subsequent stages of analysis, including sentence segmentation, LLM-based bias labelling, and dimensionality reduction.

# Chapter 6

# Prompt Engineering for Subjective Bias Annotation

## 6.1 Rationale for Prompt-Based Annotation

Subjective bias is not directly observable through keywords alone; it often appears through reasoning patterns, tone, comparison structures, or implicit expectations. Therefore, a simple rule-based approach would fail to capture these nuances. To ensure a consistent and structured annotation process, we designed a bias-aware LLM prompt capable of interpreting both lexical and semantic cues.

The prompt was structured to enforce:

- Single-label output per sentence
- Priority-based decision making (if two labels apply, choose the earlier one)
- Justification requirement—LLM must reason briefly for each decision
- Controlled JSON output for easier post-processing and validation

This design allowed LLMs to perform deterministic classification while maintaining semantic sensitivity.

## 6.2 Core Prompt Structure

The final prompt followed an *instruction* → *definitions* → *decision rules* → *input sentence* flow. An excerpt of the function used in the pipeline is shown below:

```
def build_prompt(sentence: str) -> str:
    return f"""
You␣are␣a␣professional␣annotator␣trained␣to␣identify␣bias␣
   types␣in␣media␣or␣film␣reviews.
Classify␣each␣input␣sentence␣into␣exactly␣one␣of␣the␣
   following␣nine␣categories.
Use␣both␣keyword␣cues␣and␣semantic␣patterns␣to␣justify␣your␣
   decision.
If␣multiple␣categories␣seem␣relevant,␣choose␣the␣one␣that␣
   shows␣the␣strongest␣lexical␣signal.
If␣truly␣ambiguous,␣output␣"None".

Priority␣Rule:
If␣two␣categories␣are␣equally␣possible,␣prefer␣the␣one␣
   appearing␣earlier␣in␣this␣list.

Output␣Format:
{{"label":␣"<label>",␣"reason":␣"<short reason>",␣"
   confidence":"<confidence score>"}}

The␣Bias␣Categories␣are␣listed␣below:
1.␣Expectation␣Bias␣-␣contrasts␣expectation␣vs␣reality
2.␣Comparative␣Lens␣-␣comparison␣with␣another␣work␣or␣
   standard
3.␣Representation␣Critique␣-␣evaluation␣of␣inclusion,␣
   fairness,␣casting
4.␣Omission␣Focus␣-␣absence␣of␣expected␣elements
5.␣Moral/Political␣Projection␣-␣ideological␣or␣moral␣lens
6.␣Selective␣Emphasis␣Bias␣-␣focus␣on␣one␣aspect,␣ignoring␣
   context
7.␣Performance␣Appraisal␣-␣discussion␣of␣acting/portrayal
8.␣Authenticity␣Bias␣-␣accuracy␣or␣factual␣realism
9.␣Disposition␣Bias␣-␣judgment␣driven␣by␣personal␣preference

Sentence:␣"{sentence}"
""".strip()
```

## 6.3 Prompt Refinement and Stability

Multiple iterations were tested, including:

- Adding keyword cue lists under each bias for stronger signal matching;
- Enforcing reason-before-label instruction to reduce hallucination;
- Introducing priority ordering to resolve ambiguity;
- Suggesting multilingual translation before classification for non-English sentences.

These refinements were inspired by structured prompt design principles and chain-of-thought prompting strategies in recent LLM literature.

## 6.4 Significance of Prompt Design

The prompt served as both a linguistic framework and a cognitive scaffold, enabling the LLM to infer human-style reasoning instead of mere keyword matching. This was essential to:

- Achieve reviewer-level psychological profiling later via PCA and Factor Analysis;
- Maintain consistent taxonomy usage across multiple movies;
- Enable scalable sentence-by-sentence annotation for large datasets.

Thus, the prompt functioned as a bridge between raw text and psychological inference, enabling structured analysis of human subjectivity using machine models.

# Chapter 7

# LLM-Based Bias Annotation

The extracted review corpus was annotated using Large Language Models (LLMs) to assign a dominant subjective bias label to each sentence. Since subjective bias involves implicit reasoning rather than explicit sentiment, rule-based methods were insufficient. Therefore, we employed a structured prompt-based annotation framework using LLMs to infer subjective cues from sentences.

## 7.1 Annotation Methodology

Each review was first cleaned, sentence-split, and then passed to an LLM for classification. The annotation pipeline consisted of the following steps:

1. **Text Cleaning** – Removal of HTML tags, URLs, and redundant spacing.
2. **Sentence Segmentation** – Reviews were split using `nltk.sent_tokenize()` into meaningful sentence units.
3. **Prompt Construction** – For each sentence, a carefully designed instruction prompt was generated (detailed in Chapter 6).
4. **LLM Classification** – Each sentence was passed to a language model which returned one dominant bias label.
5. **Aggregation** – Sentence-level labels were stored and later combined to compute reviewer-level tendencies.

This process is illustrated in the following script excerpt:

```
df = pd.read_csv("merged_reviews.csv")
df["sentence_labels"] = df["review"].apply(
    annotate_bias_sentences)
df.to_csv("llm_bias_labels.csv", index=False)
print("Annotated␣reviews␣saved␣to␣llm_bias_labels.csv")
```

The core of this method is the `annotate_bias_sentences()` function, which applies cleansing, sentence splitting, and model inference:

```
def annotate_bias_sentences(text: str) -> list[str]:
    if not isinstance(text, str) or not text.strip():
        return []
    text_clean = clean_text(text)
    sentences = sent_tokenize(text_clean) or [text_clean]
    labels: list[str] = []
    for sent in sentences:
        prompt = build_prompt(sent)
        result = call_llm(prompt)
        if isinstance(result, dict) and "label" in result:
            labels.append(result["label"])
    return labels
```

## 7.2 Models Used: OpenAI vs. Ollama

We experimented with two different LLM frameworks during annotation as shown in Table 7.1.

Table 7.1: Comparison of LLM models for bias annotation

| **Model Type** | **Performance** | **Notes** |
|---|---|---|
| Ollama (Llama-3, Mistral) | Moderate | Strong on simpler biases but struggled with nua |
| OpenAI (GPT-4o-mini) | Best | More consistent, lower hallucination rate |

While Ollama provided the advantage of full local execution, we observed that it occasionally failed to detect subtle psychological cues and tended to overpredict generic labels (e.g., "Expectation Bias"). OpenAI's GPT-based models showed greater precision and stability, particularly when differentiating between closely related bias types.

Therefore, while both models were tested, OpenAI models were used for the final annotation pipeline due to superior label consistency and alignment with definitions.

## 7.3 Justification for Using LLMs

Traditional NLP methods—such as keyword matching or sentiment polarity—are insufficient for modelling latent psychological constructs, such as expectation, comparison, moral judgment, or role-based performance assessment. LLMs offer the ability to infer meaning beyond surface-level vocabulary, enabling scalable annotation of subjective reasoning patterns.

Therefore, the use of LLMs allows us to approximate human-level interpretation, while maintaining computational scalability for large datasets.

# Chapter 8

# Principal Component Analysis (PCA)

## 8.1 Objective of PCA

Principal Component Analysis (PCA) was applied to understand how subjective biases vary across reviewers and to detect dominant psychological dimensions underlying reviewer behavior. Unlike Factor Analysis (discussed in Chapter 9), which focuses on uncovering latent constructs, PCA emphasizes variance maximization, helping us identify the strongest directions of subjective variation across the dataset.

The input matrix for PCA consisted of reviewer-wise bias frequency vectors, where each reviewer was represented by the distribution of the nine bias types across their reviews. Concretely, for each reviewer we count how many times each of the following labels appears in their sentences:

- Authenticity Bias
- Comparative Lens
- Disposition Bias
- Expectation Bias
- Moral or Political Projection
- Omission Focus
- Performance Appraisal
- Representation Critique
- Selective Emphasis Bias

Stacking these vectors row-wise yields a reviewer–bias frequency matrix with one row per reviewer and nine columns corresponding to the bias categories. This high-dimensional representation serves as the starting point for our dimensionality reduction.

Since the full matrix contains more than 1500 reviewers and is impractical to visualize directly, Table 8.1 presents a small excerpt to illustrate its structure.

| Reviewer ID | Auth. | Comp. | Disp. | Expect. | Moral/Pol. | Omit. | Perf. | Repr. | Sel. Emph. |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 3 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 2 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 2 | 1 |
| 3 | 2 | 1 | 6 | 0 | 5 | 0 | 0 | 3 | 0 |
| 4 | 5 | 1 | 2 | 1 | 1 | 0 | 1 | 0 | 0 |
| 5 | 0 | 1 | 1 | 1 | 2 | 1 | 3 | 1 | 2 |

Table 8.1: Excerpt of the reviewer–bias frequency matrix used as input to PCA.

Applying PCA to this matrix yields a compressed representation of reviewer subjectivity, enabling both visualization and interpretation of large-scale patterns in bias usage.

Figure 8.1 shows the projection of all reviewers into this two-dimensional PCA space, with PC1 on the horizontal axis and PC2 on the vertical axis.

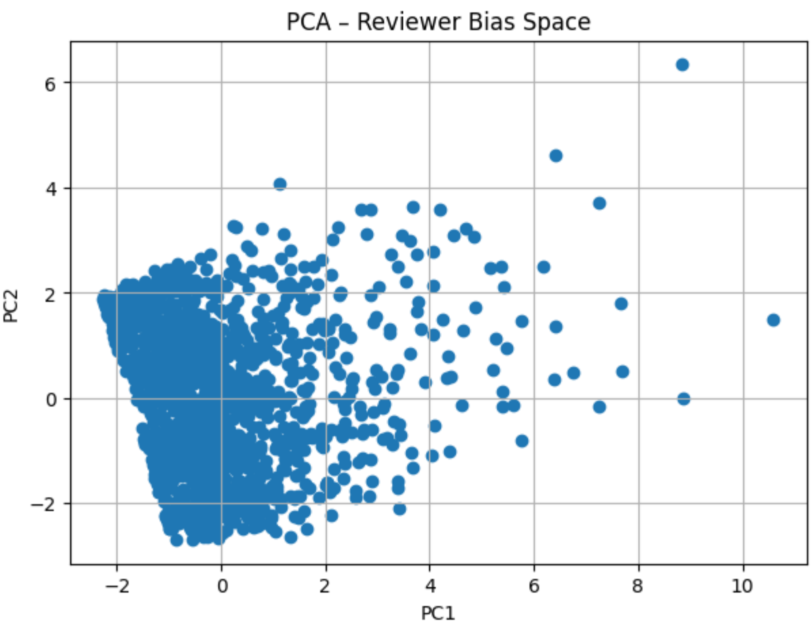


Figure 8.1: 2D PCA projection of reviewers. PC1 reflects overall subjectivity intensity, while PC2 captures the orientation of subjectivity along an emotional–moral vs analytical spectrum. Each point corresponds to one reviewer.

## 8.2 PCA Dimensions and Interpretation

Based on explained-variance thresholds and scree plot inspection, we retain two principal components. These two components capture the dominant patterns of subjective reasoning among reviewers and form the basis for our psychological interpretation.

### 8.2.1 PC1 — Overall Subjectivity / Bias Intensity

PC1 represents the degree of subjective intervention in a reviewer's writing. Reviewers with high PC1 scores frequently employ several biases simultaneously, such as:

- Omission Focus
- Selective Emphasis Bias
- Comparative Lens
- Expectation Bias
- Representation Critique
- Performance Appraisal

High-PC1 reviewers tend to produce dense and multi-layered evaluations, suggesting deeper scrutiny and critical engagement with the film. Low PC1 values instead indicate minimal or inverse subjectivity, often resembling objective reporting or neutral commentary. In the PCA plot (Figure 8.1), such reviewers appear towards the left side of the PC1 axis.

### 8.2.2 PC2 — Emotional–Moral Orientation vs Analytical Orientation

PC2 captures the *style* of subjectivity rather than its intensity. It reflects a psychological polarity between emotional–moral reasoning and analytical–technical reasoning, as summarized in Table 8.2.

Positive PC2 scores correspond to reviewers who foreground values, morals, and personal experiences, often invoking Moral or Political Projection and selective focus on particular themes. Negative PC2 scores correspond to reviewers who focus on structure, expectations, and performance, leaning towards more analytical and technically framed critiques. In Figure 8.1, this spectrum is visible along the vertical axis.

Table 8.2: Interpretation of the second principal component (PC2) as a spectrum between emotional–moral and analytical orientations.

| Positive PC2 | Negative PC2 |
|---|---|
| Emotion-driven & value-oriented<br>Emphasis on moral judgement<br>Biases: Moral/Political Projection, Selective Emphasis, Omission Focus | Analytical & expectation-based<br>Focus on technical critique<br>Biases: Expectation, Performance, Disposition |

## 8.3 Reviewer Profiling Based on PCA Coordinates

Using the PCA scores, we identified distinct reviewer categories based on their extreme positions along PC1 and PC2 in Figure 8.1.

### 8.3.1 Highly Subjective Reviewers (Top PC1)

These reviewers express multiple biases strongly and simultaneously, reflecting dense subjective intervention.

Example reviewers include:

- Reviewer 247.0: "The reviewer gives a scathing critique of the *Lincoln* film, arguing that despite Daniel Day-Lewis's exceptional acting, the movie is weighed down by a dull script and monotonous monologues. They praise Day-Lewis for delivering a magnificent performance under uninspiring circumstances, but find the film itself boring, pompous, and overly long. Instead of being inspired by a rich historical era, the reviewer feels disengaged and even wishes for a more fantastical alternative like *Abraham Lincoln: Vampire Hunter*.

  They mock the film's historical presentation—particularly the unrealistic perfection of characters' teeth—and feel that Spielberg's direction is unusually lifeless, as though he prioritized seriousness over storytelling magic. Although the film depicts political maneuvering well, the reviewer believes Spielberg was constrained by the need to impose excessive gravitas. Ultimately, they found the experience tedious and disappointing.

  The bias list highlights patterns in the reviewer's criticism, including Performance Appraisal (praising Daniel Day-Lewis while condemning the rest), Expectation Bias (judging the film against what Spielberg "should" have delivered), Authenticity Bias (complaints about unrealistic teeth and setting details), Comparative Lens (contrasting the film with more entertaining alternatives), and Moral or Political Projection (extreme hyperbole such as referencing John Wilkes Booth for shock value). These biases help explain the strongly negative and sarcastic tone of the review."

- Reviewer 82.0: high subjectivity but relatively neutral emotional tone.
- Reviewer 1382.0: highly subjective and highly emotional.

These reviewers cluster towards the far right of PC1 in the PCA space.

### 8.3.2 Oppositional / Counter-Bias Reviewers (Lowest PC1)

These reviewers display inverse or skeptical bias patterns, often challenging common interpretations or prevailing sentiments.

Examples:

- Reviewer 1483.0: "The reviewer delivers a highly negative critique of *A Beautiful Mind*, arguing that the film is historically inaccurate, emotionally manipulative, and artistically weak. They highlight major factual distortions—such as fabricating Nash's Nobel speech, downplaying the nature of the Economics Prize, and omitting Nash's bisexuality, which they see as crucial to understanding his treatment during the Cold War. They also criticize the film for presenting schizophrenia through a sentimental Hollywood lens, ignoring real medical explanations and political context.

  The reviewer condemns Ron Howard's direction as shallow and commercial, claiming he lacks insight into academia, mental illness, or the real complexities of Nash's life. They also find the performances overrated: Jennifer Connelly's portrayal is seen as historically inauthentic and shaped by Hollywood glamour, while Russell Crowe's performance is described as one-note, overly mannered, and lacking character development. The reviewer argues that other actors, such as Tom Hanks or Gary Oldman, could have delivered a more nuanced portrayal. Overall, they believe the film reduces both genius and mental illness to caricature.

  The accompanying list of biases reflects recurring patterns in the reviewer's arguments, such as Authenticity Bias (expecting strict historical accuracy), Omission Focus (criticizing what the film leaves out), Expectation Bias (judging the film against what they feel it should have been), Performance Appraisal (strong opinions on acting quality), Representation Critique (concerns about portrayals of sexuality and mental illness), and Moral or Political

Projection (linking omissions to broader social or political issues). These biases help explain the strongly opinionated and uncompromising tone of the review."

- Reviewer 1470.0.
- Reviewer 1481.0.

They occupy the far left of the PCA plot, reflecting low or oppositional bias intensity relative to the global trend.

### 8.3.3 Emotionally Dominant Reviewers (Highest PC2)

These reviewers strongly embed morals, values, emotions, and personal experiences in their reviews.

Examples:

- Reviewer 1382.0: "The reviewer praises *Vice* for Adam McKay's signature style, its humor, and especially Christian Bale's transformative performance. They note the film's political slant—portraying Dick Cheney largely as a villain—while still showing surprising moments of humanity. Although the early part of the movie drags, the reviewer finds it entertaining, well-acted, and boldly political, recommending it mainly to left-leaning audiences or fans of McKay's work.

  The bias list reflects this perspective: Expectation Bias (comparing it to *The Big Short* and McKay's past films), Performance Appraisal (heavy focus on acting), Moral/Political Projection (interpreting the film through ideological viewpoints), Disposition Bias (liking or disliking characters based on politics), and Selective Emphasis (highlighting aspects that align with the reviewer's political assumptions)."

- Reviewer 1450.0, 1464.0: strong affective projection.

They appear in the upper region of the PCA projection (high PC2), combining strong moral framing with selective emphasis.

### 8.3.4 Analytical Reviewers (Lowest PC2)

These reviewers adopt a structured, evaluative, and technical approach with minimal emotional bias.

Examples:

- Reviewer 7.0: "The reviewer strongly dislikes *Vice*, arguing that Adam McKay's comedic style—fourth-wall breaks, flashy visuals, constant narration, and tonal shifts—ruins what could have been a compelling biopic about Dick Cheney. They find the film unfocused, pretentious, overexplained, and emotionally inconsistent, especially criticizing the ending for suddenly softening its portrayal of Cheney after depicting him as monstrous throughout.

  While they think the script and direction are weak, they praise the performances, especially Christian Bale's transformation, calling it the film's only real strength. Ultimately, they see *Vice* as a wasted opportunity.

  The bias list reflects this perspective: Expectation Bias (judging McKay against what the reviewer thinks a biopic should be), Disposition Bias (strong dislike of McKay influencing the review), Performance Appraisal (heavy focus on praising the cast), Omission Focus (complaints about underdeveloped parts of Cheney's life), Moral/Political Projection (frustration with the film's treatment of Cheney's morality), and Selective Emphasis (highlighting flaws that confirm their negative view)."

- Reviewer 24.0, 8.0: highly analytical and expectation-focused.

They are located towards the bottom of the PC2 axis, where reasoning is dominated by technical evaluation, expectations, and performance rather than moral or emotional framing.

## 8.4 Summary of PCA Findings

PCA shows that subjectivity is not random, but systematically structured along two major dimensions:

1. **Intensity of Subjectivity** (PC1) — how many and how strongly biases appear in a reviewer's writing.

2. **Orientation of Subjectivity** (PC2) — whether reasoning is emotional–moral or analytical–logical.

Together with the reviewer–bias matrix structure in Table 8.1 and the PCA projection in Figure 8.1, these results indicate that reviewers tend to adopt distinct cognitive reviewing styles. These styles can be mapped, quantified, and compared computationally, forming the basis for the deeper psychological analysis and factor-analytic modelling discussed in the next chapter.

# Chapter 9

# Factor Analysis of Subjective Bias Patterns

## 9.1 Objective of Factor Analysis

While PCA helped reduce dimensionality and visualize overall bias variance, it did not fully reveal latent psychological structures underlying reviewer behavior. Therefore, Factor Analysis (FA) was applied to uncover hidden subjective dimensions that explain bias co-occurrence patterns across reviewers. This allows us to interpret how different biases cluster together and what underlying cognitive styles they represent.

## 9.2 Methodology

The dataset was converted into a reviewer–bias frequency matrix, where each row represented a reviewer and each column represented the frequency of a specific bias type. FA was conducted with varimax rotation, and the number of factors was determined based on eigenvalue $> 1$ and scree plot inspection. Two factors were retained, explaining the largest share of variance in bias usage.

Each reviewer was then mapped into a latent psychological space using their factor scores. This yielded a two-dimensional *Reviewer Subjective Map*, where Factor 1 is plotted on the horizontal axis and Factor 2 on the vertical axis. Reviewer IDs (or indices) can be overlaid on the points to retain interpretability at the individual level.

The resulting configuration, shown in Figure 9.1, exhibits a dense central region corresponding to neutral reviewers with balanced subjectivity, surrounded by sparse high-distance outliers that correspond to strongly biased reviewers.

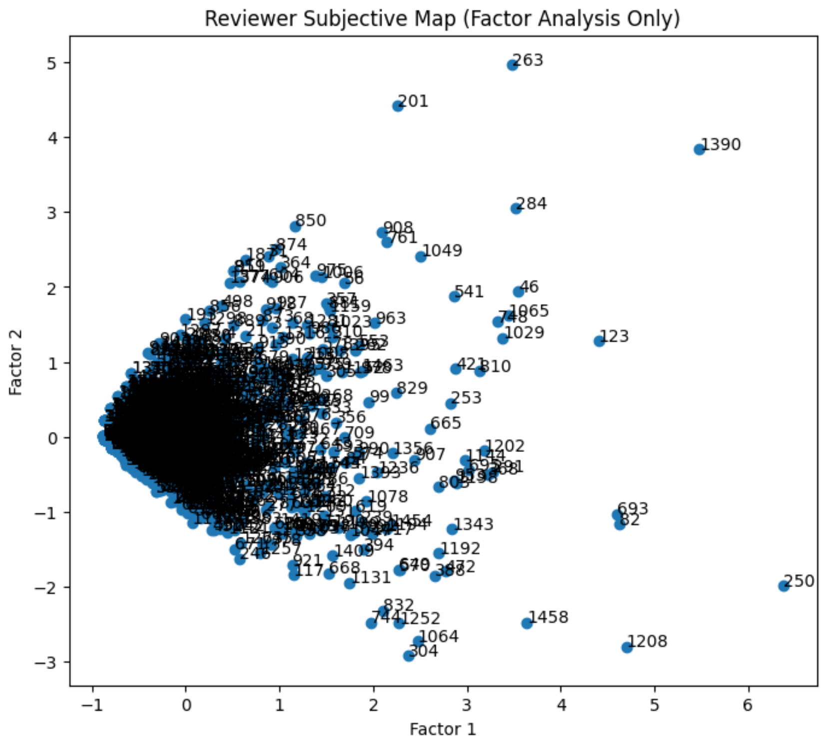


Figure 9.1: Factor Analysis projection of reviewers into latent psychological space. Factor 1 captures overall bias intensity / multi-bias engagement, while Factor 2 captures a moral–authenticity vs critical–disposition spectrum. Each point corresponds to a single reviewer.

## 9.3 Factor Loadings and Interpretation

Two major latent factors emerged with meaningful psychological interpretation.

### 9.3.1 Factor 1 — "Bias Intensity / Multi-Bias Engagement"

Factor 1 captures the overall strength and diversity of subjective interventions. High scores on this factor indicate:

- Frequent use of multiple bias types simultaneously;
- Particularly high loadings in Comparative Lens, Omission Focus, Performance Appraisal, and Selective Emphasis Bias;
- Reviewers who scrutinize many aspects of a film rather than evaluating a single dimension.

Low Factor 1 values correspond to minimal subjectivity, indicating objective or neutral reviewing tendencies. In Figure 9.1, such reviewers lie near the vertical axis around the origin, whereas strongly subjective reviewers appear far to the right.

### 9.3.2 Factor 2 — "Moral–Authenticity vs Critical–Disposition Spectrum"

Factor 2 represents a psychological polarity between two distinct reviewer styles. A concise interpretation of the two poles is summarized in Table 9.1.

Table 9.1: Interpretation of Factor 2 as a spectrum between value-driven and preference-driven reasoning.

| **Positive Factor 2** | **Negative Factor 2** |
|---|---|
| Value-driven reasoning | Preference-driven reasoning |
| Emphasis on ethical / moral concerns | Emphasis on personal likes & dislikes |
| Focus on accuracy, authenticity & fairness | Judgmental and expectation-heavy evaluations |
| Higher scores in Authenticity Bias, Moral/Political Projection, Representation Critique | Higher scores in Disposition Bias, Expectation Bias |

This reveals a cognitive spectrum: *Do reviewers judge films primarily based on objective values and fairness, or based on their own preferences and expectations?* In the factor-space plot (Figure 9.1), value-oriented reviewers occupy the upper half (high Factor 2), while preference- and expectation-driven reviewers lie in the lower half (low Factor 2).

## 9.4 Identification of Extreme and Neutral Reviewers

By computing the Euclidean distance from the origin in the factor space, we identified two broad reviewer types.

### 9.4.1 Extreme Reviewers — High Subjective Bias

These reviewers lie far from the center in Factor space and exhibit high intensity in multiple biases simultaneously. Such reviewers tend to:

- Compare films aggressively;
- Emphasize selective elements while downplaying context;

- Project strong moral or political views;
- Omit contextual information to strengthen their arguments.

These outliers represent strongly opinionated critics with distinct cognitive patterns, clearly separated from the main cluster in Figure 9.1.

### 9.4.2 Neutral Reviewers — Baseline Subjectivity

Reviewers near the origin demonstrate very low bias engagement, with no single dominant category. They reflect:

- Balanced evaluation across aspects of the film;
- Low emotional interference;
- Minimal evaluative distortion.

These reviewers form the dense core of the Reviewer Subjective Map, serving as a baseline against which more extreme cognitive styles can be compared.

## 9.5 Summary of Insights

Factor Analysis reveals that subjectivity in reviews is not random, but structured across two stable psychological dimensions:

1. **Bias Intensity / Multi-Bias Engagement** — how strongly a reviewer engages in multiple subjective biases simultaneously;
2. **Moral–Authenticity vs Critical–Disposition Orientation** — whether their reasoning is grounded in values, authenticity, and fairness, or driven by preferences, expectations, and dispositional judgments.

Together with the PCA results, the factor-space representation in Figure 9.1 supports the claim that subjective bias is a latent psychological construct that can be quantified, visualized, and analyzed computationally, rather than treated as unstructured noise.

# Chapter 10

# Results and Analysis

The analysis reveals that subjective biases are consistent, quantifiable, and psychologically structured across reviewers. By labeling sentences using LLM-based annotation and reducing dimensionality using PCA and Factor Analysis, we were able to interpret the underlying cognitive styles that drive subjective judgment in movie reviews.

## 10.1 Bias Distribution Among Reviewers

Sentence-level labels aggregated per reviewer showed that most individuals exhibit multiple biases rather than a single dominant category. Some reviewers tilted strongly toward expectation or disposition bias, while others consistently used moral or authenticity-based reasoning—indicating reviewer-specific cognitive preferences rather than pure randomness.

## 10.2 PCA Findings — Structure of Subjectivity

PCA revealed two core dimensions as summarized in Table 10.1.

Table 10.1: Summary of PCA dimensions and their interpretations

| **PCA Dimension** | **Interpretation** |
|---|---|
| PC1 | Overall subjectivity / strength of bias usage |
| PC2 | Emotional–Moral vs Analytical–Technical orientation |

High PC1 reviewers displayed dense multi-bias expression, indicating deeper interpretive scrutiny and more layered criticism. PC2 separated reviewers into value-driven (moral/emotional) and logic-driven (technical/analytical) styles, suggesting that reviewers follow distinct psychological orientations rather than random opinion patterns.

## 10.3 Factor Analysis — Latent Psychological Dimensions

Factor Analysis confirmed the PCA-derived cognitive patterns and revealed two latent constructs as shown in Table 10.2.

Table 10.2: Summary of Factor Analysis dimensions

| **Factor** | **Interpretation** |
|---|---|
| Factor 1 | Multi-bias engagement / Subjective Intensity |
| Factor 2 | Moral–Authenticity vs Critical–Disposition Spectrum |

Together, these factors indicate that reviewers differ both in *how much* they rely on subjective interventions and in *which psychological lens* they prefer when evaluating films.

## 10.4 Reviewer Categories Identified

By computing distance from the factor origin, reviewers could be classified into four psychological categories as described in Table 10.3.

Table 10.3: Reviewer categories identified through factor analysis

| **Reviewer Type** | **Description** |
|---|---|
| Extreme Subjective | Strong multi-bias patterns, deep personal reasoning; highly opinionated critics. |
| Oppositional Reviewers | Inverse/breakdown bias patterns, skeptical tone; challenge dominant narratives. |
| Emotional Reviewers | Morally expressive and narrative-driven; foreground values and experiences. |
| Analytical Reviewers | Technical, expectation-based, detached reasoning; focus on structure and craft. |

These categories demonstrate that subjectivity is structured along interpretable reviewer "personas" rather than being diffused noise.

# Chapter 11

# Conclusion and Future Work

## 11.1 Summary

This study demonstrates that subjectivity in movie reviews is computationally detectable and psychologically meaningful. By constructing a taxonomy of nine bias categories and using LLMs for sentence-level annotation, we quantified subjective expression across thousands of reviews. PCA and Factor Analysis further showed that:

- Subjective interventions follow distinct cognitive orientations, not random variability.
- Reviewers exhibit interpretive identities, measurable across latent psychological dimensions.
- Bias intensity and expression style can be mapped and visualized, allowing clustering of reviewers by mindset rather than sentiment alone.
- LLMs—especially GPT-based—provide a reliable approximation of human interpretation when guided by carefully designed prompts.

## 11.2 Key Contributions

This project makes the following contributions to the field:

1. A formal taxonomy of subjective biases in reviews.
2. A scalable LLM-based annotation pipeline for sentence-level bias labelling.
3. Statistical modeling of subjectivity using PCA and Factor Analysis.
4. Visualization of reviewer psychology along latent subjectivity dimensions.

## 11.3 Limitations

While our approach demonstrates promising results, several limitations should be acknowledged:

- The taxonomy is limited to nine bias types, which may not capture all forms of subjective intervention.
- LLM annotation, while effective, may introduce systematic biases based on the model's training data and prompt design.
- The study focuses exclusively on movie reviews, limiting generalizability to other domains such as product reviews or political discourse.
- Sarcasm and irony detection, while theoretically motivated through salience, may still pose practical challenges.

## 11.4 Future Directions

Several promising directions for future work include:

- **Multimodal Analysis:** Explore multimodal subjectivity by incorporating audio/visual cues from films.
- **Improved Irony Detection:** Develop more sophisticated methods for handling sarcasm and irony in reviews.
- **Hybrid Models:** Combine CRF, LSTM, or transformer-based classifiers with LLM cues for improved accuracy.
- **Domain Expansion:** Extend the framework to political debates, news articles, and product reviews.
- **Real-Time Classification:** Train a fine-tuned classifier for real-time bias detection in streaming review data.
- **Cross-Cultural Analysis:** Investigate how bias patterns vary across different cultural and linguistic contexts.
- **Temporal Dynamics:** Analyze how reviewer bias profiles evolve over time for long-term reviewers.

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